Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/29/2021
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Exploring Spaceflight-Linked Changes in Plant Defense Capabilities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2022
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
Contact Email:	gioia.massa@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2022 per NSSC information (Ed., 1/4/22) NOTE: End date changed to 10/25/2021 per G. Massa/KSC; forthcoming in NSSC (Ed., 9/24/20)		
Key Personnel Changes/Previous PI:	December 2021 report: None.		
COI Name (Institution):	Swanson, Sarah Ph.D. (University of Wisconsin, Madison)		
Grant/Contract No.:	80NSSC19K0126		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Analysis of the plant biology transcriptomics and proteomics data in the GeneLab data system reveals that a common feature of spaceflight is alteration in plant defense-related transcripts and proteins in space. Anecdotally, plants have also been seen to suffer from pathogen attack during spaceflight missions. Taken together, these observations suggest a hypothesis that plant defense responses may be compromised by spaceflight. Such a disruption of defense would have important implications for the management of plants grown as part of an astronaut life support system. This is especially relevant given the large microbial community on board the International Space Station (ISS) and the possibility that spaceflight triggers enhanced microbial pathogenicity. However, the critical unanswered question is whether the altered defense gene expression seen in spaceflight truly plays out as altered susceptibility to pathogens. Use of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware growing Arabidopsis seedlings allows direct testing of this idea. Specifically, the project will: (1) Capitalize on the unique capabilities of the BRIC-LED hardware to analyze the transcriptome and cell wall structure responses of Arabidopsis seedlings grown in spaceflight. Plants will be grown in the hardware and then challenged with the bacterial elicitor peptide flg22. Comparison of subsequent on-orbit responses in wild-type and mutants in the defense regulators AtRBOHD and CML24 will then be made. (2) Use comparisons to ground-based control experiments and clinostat and hypoxia treatment to ask whether such changes are likely linked to the microgravity or hypoxic elements of spaceflight and whether these changes may affect the ability of the plant to defend itself against pathogen challenge. The BRIC-LED/Petri Dish Fixation Unit (PDFU) hardware is well-suited to this experimental approach. Thus, the two-stage PDFU hardware will allow both stimulation (injection of flg22 elicitor onto the seedlings) and subsequent fixation of samples (with RNAlater). As the experiment would use flg22 peptide to elicit defense response, potential problems with flying live pathogen will be circumvented. The experimental design will capitalize on the minimal crew time needed for a BRIC-based experiment and the unique capabilities of the BRIC hardware to treat plants in situ on orbit. The results from this experiment will allow direct testing of the critical question of whether the ability of plants to mount a defense response is compromised during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how pathogen sensing and response operates in plants, targeting the Ca2+-dependent and reactive oxygen signaling machinery. The results from this work are not only providing insight into how pathogen signals impact plant growth and defense responses in space but also are contributing to our understanding of similar responses on Earth. Pathogen attack is a major limit on crop productivity with estimates of global losses in the range of 20% across multiple major crops such as wheat, maize, rice, soybean, and potato. The pathogen sensing and response systems triggered by the molecular signals of bacterial attack in space are also those employed by plants challenged by pathogens on Earth. Thus, this work aimed at analysis of how reactive oxygen species and Ca2+ signaling is involved in the defense response systems induced during spaceflight should also help our understanding for these same systems in this general plant defense response system on Earth. These results will provide information to highlight novel molecular targets for potential manipulation to tailor plant responses to the wide spectrum of pathogens that attack plants.
Task Progress:	The flight element of this project is currently awaiting availability of the BRIC-LED hardware and associated flight opportunity. The ground-based components have made progress on bioinformatics analysis and tool development. Initial bioinformatics-based analysis suggested that pathogen response might be disrupted in plants growing in spaceflight. Therefore, efforts have been underway to extend these intial bioinformatics-based analytical pipelines. In collaboration with the GeneLab plant analysis working group, a comprehensive analysis of the plant spaceflight transcriptomes deposited at the GeneLab data repository has been made, focusing on transcriptomics experiments performed using the BRIC hardware to find common BRIC-related spaceflight response signatures. This analysis has reinforced many of the expected molecular fingerprints of spaceflight, such as oxidative stress and cell wall modifications but has also revealed interesting new candidate pathways, such as parts of the plant cold response system. This BRIC-LED 002 experiment will compare wild type responses to two sets of mutants in the genes for the proteins Calmodulin-like 24 and AtRBOHD. Therefore, the bioinformatics tools outlined above have been used to ask how these genes might play roles in biological signaling related to the expected stresses in spaceflight. These analyses suggest that especially for RBOHD, mutants in this gene show altered patterns of transcription related to primary metabolism that would be consistent with a pre-adaptation to flooding and hypoxic stress. As both of these stresses are thought to be likely features of the spaceflight environment, this observation suggests RBOHD mutants are likely to be resistant to these stresses imposed when growing on orbit. Analysis has therefore been made for the flooding and hypoxia growth phenotypes of knockout mutants in both CML24 and RBOHD in ground-based assays to provide some context for possible phenotypes seen in spaceflight. These studies show that both RBOHD and CML24 are involved in the response to hypoxia and flooding challenge. This resistance is visible at levels from maintained growth to reduced induction of transcriptional markers for hypoxic stress. RBOHD is thought to produce reactive oxygen species (ROS) that signal to a suite of stress responses, suggesting that the altered flooding and hypoxic response in mutants in this gene might be linked to alteration in a wide range of other stress-related phenotypes. Phenotyping and bioinformatics-based analysis of the knockout mutant in RBOHD for responses related to wounding (a stress closely aligned to induction of pathogen response in the plant) suggest alterations in mechanical wound response systems and those related Jasmonic Acid-mediated signaling may accompany growth in space. Unexpectedly, despite this close link between ROS and wound response systems, loss of RBOHD function does not appear to impair the plant-wide damage response signaling. During 2020-2021, this work has been presented at the annual meeting of the American Society for Gravitational and Space Research, the American Society for Plant Biology, and at the ISS Science Symposium. Outreach efforts have also been ongoing within the context of the pandemic-related limitations on travel and contact with virtual presentations to journalism students at Penn State and to an international audiences at the Universidade Nova de Lisboa in Portugal and at the Symposium of the Space Generation Advisory Council in Taiwan. This project was also part of a virtual outreach program to under-represented groups in (Science, Technology, Engineering, and Mathematics), including the PEOPLE (Precollege Enrichment Opportunity Program for Learning Excellence) program and in outreach to the general public through participation in programs such as the Friends of Allen Centennial and Madison Master Gardeners. In person outreach activities were possible with school students at the Deke Slayton Memorial Space and Bike Museum in Sparta, WI.

Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Johns S, Hagihara T, Toyota M, Gilroy S. "The fast and the furious: Rapid long-range signaling in plants." Plant Physiol. 2021 Mar;185(3):694-706. https://doi.org/10.1093/plphys/kiaa098 ; PMID: 33793939; PMCID: PMC8133610 . , Mar-2021
Articles in Peer-reviewed Journals	Uemura T, Wang J, Aratani Y, Gilroy S, Toyota M. "Wide-field, real-time imaging of local and systemic wound signals in Arabidopsis." J Vis Exp. 2021 Jun 4;(172):e62114. https://doi.org/10.3791/62114 ; PMID: 34152317 , Jun-2021